

# Tom de Groot

## INDEPENDENT RESEARCHER

AI-assisted mathematics | formal verification | scientific ML

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GitHub | Research notes | LinkedIn

EXTERNAL VALIDATION

**"the gold standard of LLM-aided mathematics"**

Daniel Litt on FirstProof Problem 9

## SELECTED RESEARCH

OPEN PROBLEM | CANDIDATE PROOF

### A Raynaud-Bundle Route

Jul 2026

Steered GPT-5.5 Pro and GPT-5.6 Sol Pro from a partial result to a complete candidate proof of Daniel Litt's problem. Auditing the argument with his feedback. Research note | Proof thread

FIRSTPROOF | LEAN VERIFIED

### Problem 9

Feb 2026

Orchestrated LLM agents from informal search to a Lean 4 proof. The repository's only solved problem. Proof + code

OPEN-SOURCE FORMALIZATION

### Prime Number Theorem And (PNT+)

2025 - Present

Merged Lean proofs; compilation and adversarial review as hard checks on model-generated mathematics. Merged PRs

## SCIENTIFIC & AGENT SYSTEMS

SIMULATION | CERTIFICATION | RL

### Bazaar_ML

2026 - Present

**49,248** executable evaluation cases; **99.85%** sampling-pool coverage; deterministic traces, provenance, and agent-training hooks.

CHEMISTRY | GRAPH NEURAL NETWORKS

### NMR Prediction

2024 - 2025

Curated molecular/NMR data and benchmarked GNNs on HPC under controlled splits and evaluation.

AGENTIC ENGINEERING

### BaanSoliciteren

2025 - 2026

Auditable pipeline for vacancy ingestion, fit scoring, and state tracking. GitHub

## AT A GLANCE

**LEAN 4**
Machine-checked FirstProof solution

**49,248**
Executable evaluation cases

**99.85%**
Sampling-pool coverage

## EXPERIENCE

### AI Engineer

SoundImports | Present

Prediction systems, AI automation, and containerized AWS services.

### Earlier

Teamspot | Transcom | Chess.com

Regional direction, QA, support, and international events.

## METHODS & TOOLS

**Research**
Agent workflows, evaluation design, formal verification

**Code**
Python, Lean 4, PyTorch, SQL, RDKit

**Systems**
Linux/HPC, AWS, Docker, Git

## EDUCATION

### BSc Chemistry

University of Groningen | 2021 - 2025

Chemistry of Life. Thesis: GNN evaluation for NMR prediction.

## RECOGNITION

Harmonic Rising Mathematician

Chess.com Member of the Year 2021